Project Description: UK Train Rides Analysis

The "UK Train Rides" project is an advanced data analysis initiative aimed at exploring, analyzing, and forecasting railway transportation metrics using historical datasets. By applying data science techniques, the project translates raw passenger data into strategic insights that can help transportation authorities optimize routes, manage pricing, and improve customer satisfaction.

1. Core Objectives

- **Data Cleansing:** Establish a robust data model and perform thorough preprocessing on the `railway.csv` dataset.
- **Business Intelligence:** Identify critical patterns regarding peak travel times, popular routes, and revenue generated by different ticket classes.
- **Forecasting:** Utilize machine learning to predict the number of train rides for the upcoming month, estimated daily revenue, and expected demand across various ticket tiers.

2. Project Phases & Methodology

The project spans a four-week lifecycle, employing a modern data stack (SQL, Python, and Tableau):

- **Week 1 (Data Prep):** Building the foundation. The dataset is ingested, cleaned, and structured using SQL and Python libraries (pandas).
- **Week 2 (Exploratory Analysis):** Generating analytical questions and creating initial visualizations to uncover historical trends.
- **Week 3 (Predictive Modeling):** Applying Scikit-learn to train forecasting models, focusing on anticipating future passenger demand and financial outcomes.
- **Week 4 (Dashboarding):** Consolidating all findings into an interactive Tableau dashboard, accompanied by a comprehensive final presentation.

Deliverables: A clean dataset, well-documented Python notebooks, an interactive Tableau visualization dashboard, and this final project report.